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Deep Double Descent

Where bigger models, longer training, and more data can hurt

COMP34312 | Mathematical Topics in Machine Learning | **Group:** Ayush Kumar | Jacob Yip | Harshita Dassani

Objective Explain why test error can peak *near interpolation* and then fall again.
Problem Classical bias–variance theory predicts one U-shape; modern overparameterised nets interpolate yet generalise.
Approach Sweep width, training time, samples, noise, optimiser, regularisation; track test *and* train error.
Contribution Nakkiran et al. unify model-wise, epoch-wise, and sample-wise double descent via **Effective Model Complexity** (EMC).

1 From the textbook U-curve to the double-descent peak

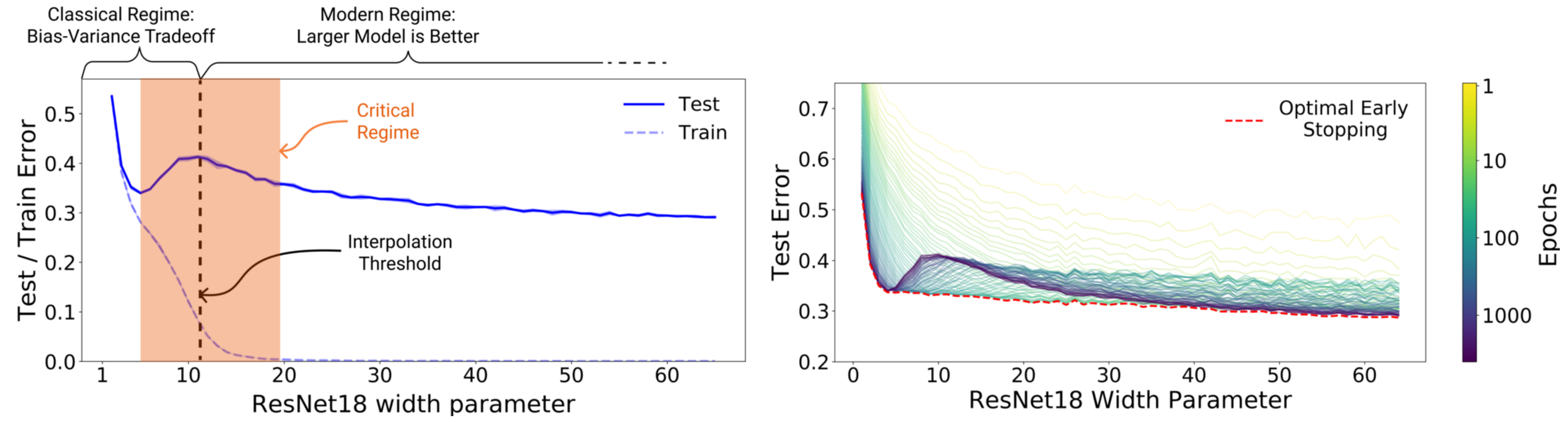

Fig. 1, Nakkiran et al. (2019). Left: test error on ResNet18s, varying width, CIFAR-10 + 15% label noise. **Right:** test error coloured by training epoch; dashed red curve denotes optimal early stopping.

THESIS
For modern training procedures the population-risk curve exhibits *two* descents, separated by a sharp peak at the **interpolation threshold** at which the procedure first fits the training set exactly.

BEYOND SECTION 5.3
The lecture introduces the phenomenon and links it to implicit regularisation. We extend this by substituting *parameter count* with **Effective Model Complexity** (EMC) – the largest train-set size on which the procedure $\mathcal{T}$ still drives training error below ε – and demonstrating that the second descent occurs **model-wise**, **epoch-wise**, and **sample-wise**, unified under the **Generalised Double Descent Hypothesis**.

HOW TO READ FIG. 1
- **Classical regime** (left of peak): U-shape; classical bias-variance trade-off applies.
- **Critical regime** (coral band): test error *spikes* near $\mathrm{EMC} \approx n$; the *interpolation threshold* sits at its centre.
- **Modern regime** (right of peak): test error *descends a second time* as capacity grows.
- **Dashed red curve** (right): optimal early stopping; tracks the descending epoch-wise envelope.

2 Setting: the lecture decomposition we extend

Following the course notation (Notes §5.1, Eqs. 68–73):

$$\hat{f} \leftarrow \mathcal{A}(\mathcal{F};\mathcal{S}),\ R(f) := \mathbb{E}_{xy}[\ell(\mathbf{y}, f(\mathbf{x}))],\ \widehat{R}(f,\mathcal{S}) := \frac{1}{n}\sum_{i=1}^{n}\ell(\mathbf{y}_i, f(\mathbf{x}_i)).$$

$\hat{f}$ is found by gradient descent (Notes §3.4):

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}}\widehat{R}(f_{\boldsymbol{\theta}_t},\mathcal{S}).$$

The excess risk decomposes into three statistically meaningful gaps:

$$\underbrace{R(\hat{f}) - R(y^*)}_{\text{excess risk}} = \underbrace{R(\hat{f}) - R(f_{\text{erm}})}_{\text{optimisation}} + \underbrace{R(f_{\text{erm}}) - R(f^*)}_{\text{estimation}} + \underbrace{R(f^*) - R(y^*)}_{\text{approximation}}.$$

The classical bias-variance trade-off bundles *approximation + part of estimation* into **bias** and the rest into **variance**; *incomplete*, not wrong. The **paper's pivot:** treat the entire procedure $\mathcal{T}$ – architecture, optimiser, training time, regularisation, and label policy – as the single object whose EMC matters.

TERM-BY-TERM BEHAVIOUR AT THE PEAK
Optimisation declines with training time and width; **estimation** *spikes* at $\mathrm{EMC} \approx n$; **approximation** declines with hypothesis-class size.

3 Definition + Hypothesis (Nakkiran et al. 2019)

DEFINITION 1: EFFECTIVE MODEL COMPLEXITY
A **training procedure** $\mathcal{T}$ maps any labelled train set $S = \{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}_{i=1}^{n}$ to a classifier $\mathcal{T}(S)$. For an error tolerance $\varepsilon > 0$,

$$\boxed{\mathrm{EMC}_{\mathcal{D},\varepsilon}(\mathcal{T}) := \max\{\, n \mid \mathbb{E}_{S\sim\mathcal{D}^n}[\mathrm{Error}_S(\mathcal{T}(S))] \le \varepsilon \,\}}$$

i.e. the largest sample size on which $\mathcal{T}$ drives train error below ε. Note that EMC depends on the optimiser, training time, augmentation, and labels, not solely on the architecture.

HYPOTHESIS 1: GENERALISED DOUBLE DESCENT
For "natural" $\mathcal{D}$, neural-network procedures $\mathcal{T}$, and small ε (paper: $\varepsilon = 0.1$):

- $\mathrm{EMC} \ll n$ **Under-parameterised:** any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.
- $\mathrm{EMC} \approx n$ **Critically parameterised:** the sign of the same change is *indeterminate*.
- $\mathrm{EMC} \gg n$ **Over-parameterised:** any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.

Why this is a hypothesis, not a theorem. ε and "sufficiently smaller / larger" are heuristic; the critical-interval width is not formally specified for deep nets. **EMC is operational, not predictive**: peak location must be *measured*, not derived.

5 Sample-wise descent: increasing n can locally raise test risk

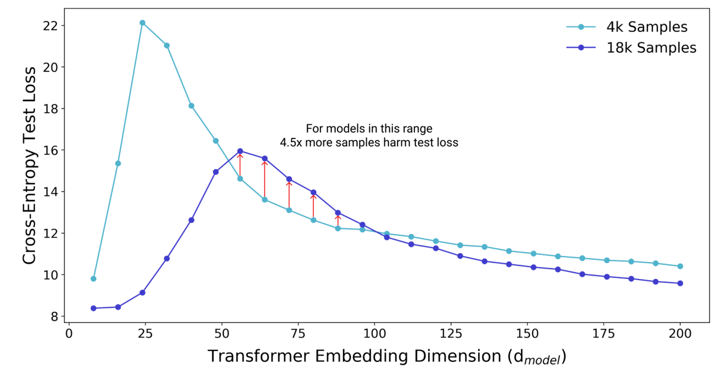

Fig. 3, Nakkiran et al. IWSLT'14 De-En, 6-layer Transformer, $d_{ff} = 4d_{\text{model}}$, 80k gradient steps, 10% label smoothing. Highlighted bars indicate the d_{model} interval over which a $4.5\times$ increase in n *raises* test loss.

- Fixing $(\mathcal{T}, \mathcal{F})$ and increasing n leaves the interpolation threshold in EMC space unchanged but shifts n relative to it; widths near the small-sample threshold are thereby displaced into the critical regime.
- The non-monotonicity is local: asymptotically, additional samples reduce excess risk.
- **Practical implication:** an **ensemble of models trained on disjoint sub-samples** sits past each member's peak; averaging cancels member-specific interpolation noise (Fig. 28).

4 Main empirical evidence: model-wise × epoch-wise double descent

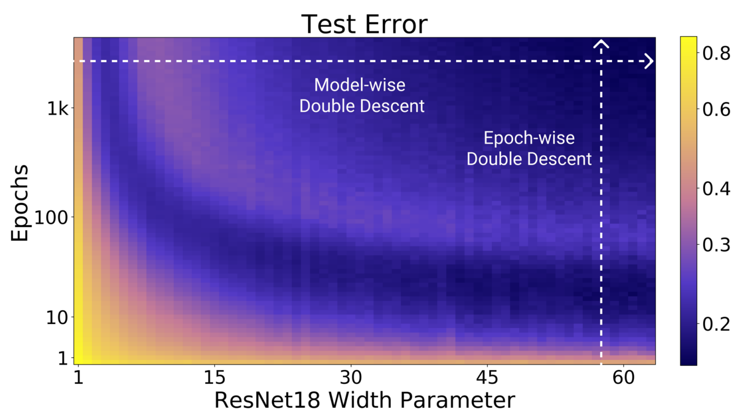

Fig. 2 (left), test error. Test error attains ≈ 0.47 at width $k \approx 10$ and decreases to ≈ 0.24 at $k = 64$.

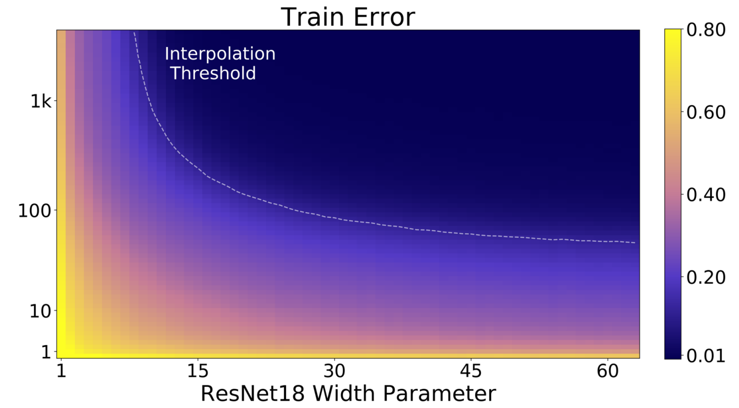

Fig. 2 (right), train error. Dashed curve denotes the interpolation threshold ($\widehat{R} \to 0$).

READING THE HEATMAPS
- **Same axes:** ResNet18 width $k \in [1, 64]$ vs training epochs (log).
- **Horizontal slice:** **model-wise** peak as width grows. **Vertical slice:** **epoch-wise** peak as training continues.
- **Key alignment:** the locus of high test error in the left heatmap coincides with the train-interpolation contour (dashed) in the right heatmap; the peak occurs *at* the threshold, not past it.

SETUP + KEY NUMBERS (PAPER §4)
ResNet18, CIFAR-10, **15% label noise**, augmentation, Adam (lr 1e-4), 4k epochs. Pattern identical under SGD, on CIFAR-100, and at **0% noise** (Fig. 4a). Sample-wise (Fig. 3): a **4.5× data increase** (4k→18k) raises test loss for $d_{\text{model}} \in [50, 100]$. Paper uses $\varepsilon = 0.1$.

The unifying variable is EMC: any modification of $\mathcal{T}$ that moves EMC across n shifts the peak correspondingly.

6 Mechanism (rigorous for linear models; conjectural for deep nets)

CRITICALLY PARAMETERISED
The hypothesis class admits effectively a unique interpolant. Constraining it to fit noisy or mis-specified labels perturbs the on-distribution map; test risk consequently spikes.

OVER-PARAMETERISED
A continuum of interpolants exists. The optimiser's **implicit bias** (e.g. SGD toward minimum-norm) selects an interpolant that absorbs label noise while preserving smoothness on the data manifold.

$\mathrm{EMC} \ll n$	$\mathrm{EMC} \approx n$	$\mathrm{EMC} \gg n$
unique interpolant	peak	continuum
classical U-curve	*critical regime*	implicit bias selects

WHERE THE MECHANISM IS PROVEN
Mei & Montanari (2022, Thm. 2) establish a test-risk peak at $d = n$ for random-features regression under mis-specification; Belkin et al. (2019) prove an analogous peak for least-squares; Bartlett et al. (2020) prove benign overfitting under minimum-norm interpolation. The deep-network analogue remains conjectural.

7 Critical reflection: what we do *not* claim

- **The bias-variance trade-off is not invalidated.** It remains correct within a fixed hypothesis class; it is *incomplete* once $\mathcal{T}$ is permitted to vary.
- **Model width, training time, and sample size are not monotonic in test risk.** Within the critical interval each can locally *increase* it.
- **Label noise is not the cause of the peak.** Noise amplifies it; the peak persists at **0% noise** (CIFAR-100, Fig. 4a; CNN, Fig. 7) under model mis-specification.
- **Parameter count does not equal EMC.** Parameters constitute one input to EMC; optimiser, epochs, augmentation, and label policy contribute jointly.

8 Practical implications and open questions

IMPLICATIONS FOR PRACTICE
- **Diagnostic.** Train and test curves decouple near the interpolation threshold; monitor both.
- **Per-model mitigation.** Weight decay, early stopping, or scaling well beyond the threshold.
- **Aggregate mitigation.** Ensemble K sub-sampled models, each over-parameterised relative to its own n (Fig. 28).
- **Augmentation.** Data augmentation shifts the peak rightward (Fig. 5) without changing the architecture.

OPEN SCIENTIFIC QUESTIONS
- A principled choice of ε and closed-form critical-interval width.
- A non-linear EMC theorem for over-parameterised networks.
- Characterisation of SGD's implicit bias for realistic architectures and losses.

Interpolation is a threshold to understand, not a final verdict.

KEY REFERENCES
[1] Nakkiran, Kaplun, Bansal, Yang, Barak, Sutskever. *Deep Double Descent.* arXiv:1912.02292 (2019). [2] Belkin, Hsu, Ma, Mandal. *Reconciling modern ML practice and the classical bias-variance trade-off.* PNAS 116(32) (2019). [3] Zhang, Bengio, Hardt, Recht, Vinyals. *Understanding deep learning requires rethinking generalization.* ICLR (2017). [4] Mei & Montanari. *The generalization error of random features regression.* Comm. Pure Appl. Math. (2022). [5] Bartlett, Long, Lugosi, Tsigler. *Benign overfitting in linear regression.* PNAS 117(48) (2020). [6] Brown & Mukherjee. *COMP34312 Notes 2026, §5.*

AI-USE DECLARATION
Generative AI was used solely as a LaTeX typesetting assistant for the poster source. All conceptual content, figure choices, mathematical claims, and source references were directed and verified by the authors. Figures 1–3 are reproduced from Nakkiran et al. (2019).